THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — SCIENCE TWO-CHAPTER TEST SET C

Chapters 1 & 2: **Exploring Substances: Acidic, Basic, and Neutral, Electricity: Circuits and Their Components** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Which of these is acidic? 1

(a) Soap solution **(b)** Lime water
(c) Baking soda solution **(d)** Lemon juice

2. The acid present in vinegar is: 1

(a) citric acid **(b)** acetic acid **(c)** formic acid **(d)** tartaric acid

3. The pain of an ant bite is due to: 1

(a) citric acid **(b)** carbonic acid **(c)** formic acid **(d)** acetic acid

4. Red rose extract (China rose) turns ___ in acids. 1

(a) yellow **(b)** green
(c) dark pink or magenta **(d)** blue

5. Antacids such as milk of magnesia are: 1

(a) basic **(b)** sour **(c)** acidic **(d)** neutral

6. The path along which electric current flows is called an electric: 1

(a) circuit **(b)** filament **(c)** cell **(d)** conductor

7. The symbol of a cell is drawn with: 1

(a) two equal lines **(b)** a long line and a short line
(c) a zigzag **(d)** a circle

8. Which of the following is a conductor? 1

(a) Rubber **(b)** Copper wire **(c)** Plastic scale **(d)** Dry wood

9. The metal used in wires for good conduction is mostly: 1

(a) wood **(b)** copper **(c)** rubber **(d)** glass

10. The electricity from the power plant at home is: 1

(a) direct current only **(b)** static charge
(c) alternating current (AC) **(d)** solar current

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 15–25 words each.

11. Why does a factory treat its acidic waste before releasing it? 2

12. How can we make a paper indicator from turmeric? 2

13. Why does the LED glow only in one connection? 2

14. State whether a graphite pencil lead conducts. Why is that unusual? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. What is the use of an indicator in a neutralisation reaction? 3

16. Draw the symbols (describe) of a bulb, battery and switch used in circuit diagrams. 3

17. Why is Nature called our science laboratory? Give two examples of natural indicators. 3

OR

List three differences between a neutral and an acidic solution. 3

18. A circuit has a cell, a bulb, a switch and a rubber band joining two wires. Does the bulb glow? Explain. 3

OR

What is the difference between a conductor and an insulator in terms of use in cables? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 80–120 words each.

19. (i) Why do chefs add lemon juice to some dishes? 5
(ii) Why do bakers use baking soda?
(iii) What do these show about acids and bases?

OR

Case study: Four bottles P, Q, R, S contain lemon juice, soap solution, sugar solution, vinegar. Litmus results: P turns blue to red, Q turns red to blue, R no change, S turns blue to red. 5
(i) Identify.
(ii) Which two are acids?
(iii) Which is neutral?
(iv) Which is a base?

20. (i) Explain how a lamp can be made to glow brighter using two cells. 5
(ii) Draw the diagram (describe).
(iii) What is the limit?

OR

Case study: A town's power lines fall during a storm. 5
(i) Why is it dangerous to go near them?
(ii) What is the type of current?
(iii) Which materials are safe to use?
(iv) Whom to inform?

END OF QUESTION PAPER — **Total: 40 Marks**